
MEvD-GRN: Multi-Evidence Distillation for Gene Regulatory Network Inference via Curriculum Learning over Confidence-Tiered Evidence

MEvD-GRN Project ¹

Abstract

Inferring gene regulatory networks (GRNs) – the directed map from transcription factors (TFs) to the target genes they control – is a central problem in computational biology, and single-cell multi-omic profiling (paired RNA and ATAC sequencing) is rapidly becoming its primary data substrate. Existing reference databases seldom acknowledge that not all regulatory edges are equally trustworthy: an edge inferred from a TF binding assay (localization evidence) is weaker than one inferred from a functional knockout (perturbation evidence), and edges supported by *both* are the most reliable. We introduce **MEvD-GRN**, a compact dual-modality graph neural network that treats this confidence structure as a training curriculum rather than as an incidental labeling detail. MEvD-GRN encodes RNA and ATAC signals in separate towers, fuses them only at a role-aware decoder in which chromatin accessibility gates transcriptional compatibility, and is trained sequentially on the localization and perturbation evidence tiers of the K562 cell line from the SC-MO-GRN-DB resource. We then evaluate the trained model in a *zero-shot* fashion on the highest-confidence dual-evidence tier – never used for a single gradient update – and find that a 306,691-parameter model reaches AUPR 0.558 / AUROC 0.838 on this held-out tier, exceeding a 4.08M-parameter supervised multi-omic GNN baseline (scMultiomeGRN) and three additional classical and deep-learning baselines. We complement this headline result with a rigorous ablation study (10 configurations) that surfaces a genuine limitation – sequential curriculum training induces catastrophic forgetting of the localization tier, which memory replay only partially

remedies – and we quantify exactly how much of the model’s competitiveness against the strongest baseline depends on the specific curriculum schedule chosen.

1. Introduction

Every human cell shares the same genome, yet a skin cell and a neuron express wildly different sets of genes. The switches that decide, for a given cell type, *which* genes are turned on or off are transcription factors (TFs) – proteins that bind DNA near a target gene and either promote or repress its expression. The full wiring diagram of “which TF regulates which gene” in a given cell type is called its **gene regulatory network (GRN)**. Reconstructing GRNs computationally from measurable molecular data is important because it lets researchers predict the downstream effect of perturbing a single gene, understand how cell identity is maintained or lost (e.g. in cancer or reprogramming), and prioritize candidate drug targets – all without having to experimentally knock out every TF in every cell type, which is infeasible at scale.

GRN inference as link prediction. Formally, a GRN is a directed bipartite graph: one set of nodes is a subset of “regulator” genes (the TFs), the other set is all genes in the genome, and a directed edge $TF\ i \rightarrow gene\ j$ means TF i regulates gene j . GRN inference is therefore a **link prediction** problem, scored over a candidate space of size $|TFs| \times |genes|$ that is enormous (millions of candidate pairs) relative to the handful of edges that are actually true.

Two complementary experimental measurements. Modern single-cell technology provides two views of a cell’s regulatory state that are individually informative but jointly more powerful: **scRNA-seq** measures which genes are being transcribed (expression), while **scATAC-seq** measures which stretches of DNA are physically accessible to be bound by a TF at all (chromatin accessibility). A TF cannot regulate a gene whose locus is closed, and a gene cannot be considered “regulated” in a meaningful sense if it is not also transcriptionally responsive – so the two modalities constrain each other (Duren et al., 2018). Most GRN inference methods, however, still use only one of the two (usu-

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ally RNA alone); we build a model that treats accessibility and expression as functionally distinct, non-interchangeable roles (Section 5).

The overlooked structure: evidence is not created equal.

The public database we build on, SC-MO-GRN-DB (Valensi et al., 2026), is unusual in that it tags every reference regulatory edge with *how it was experimentally discovered*. This produces a natural confidence hierarchy with three tiers:

- **Localization** edges come from TF-binding assays (ChIP-seq): high recall (many binding events are detected) but low precision, because physical binding does not guarantee a functional regulatory effect.
- **Perturbation** edges come from knock-down/knockout/CRISPR experiments: low recall (only genes that respond measurably are captured) but high precision, since a causal expression change is direct functional evidence.
- **Dual-evidence** edges are the intersection of the two: both a binding event *and* a functional effect are observed for the same TF-gene pair. This is the smallest and most trustworthy tier – the closest available proxy to biological ground truth.

Almost every GRN inference method in the literature is agnostic to this structure: reference networks are pooled into a single, undifferentiated set of positive edges before training or evaluation. Our central methodological contribution is to instead treat the tier hierarchy explicitly, as an **evidence-tier curriculum**: train sequentially on localization then perturbation evidence, and hold the dual-evidence tier out entirely as a zero-shot evaluation benchmark. Section 6 explains in detail why this design – rather than directly fine-tuning on the dual-evidence tier – is the more informative test of generalization, given how the tiers are structurally related to one another.

Contributions. (1) We propose MEvD-GRN, a role-aware dual-modality GNN that keeps RNA and ATAC representations in separate encoder/GNN towers and fuses them only through an accessibility-gated decoder (Section 5). (2) We formalize an evidence-tier curriculum with a global, hierarchy-consistent edge splitting protocol that provides a provable guarantee against a model being evaluated on an edge it was trained on under a different tier (Section 4), and use this guarantee to justify zero-shot evaluation of the dual-evidence tier as the primary generalization metric (Section 6). (3) We provide a rigorous empirical comparison against four baselines – two classical/diffusion RNA-only methods and a 4.08M-parameter supervised multi-omic GNN (scMultiomeGRN (Xu et al., 2025)) – on identical splits (Section 9). (4) We run a 10-configuration ablation study that identifies catastrophic forgetting as the main open weakness of

the sequential curriculum, quantify how much of it memory replay recovers, and identify two alternative training schedules (joint multi-task training and a full replay-based curriculum) that substantially close the gap (Section 10).

We report all results honestly, including configurations under which the proposed model *loses* to a baseline, because we believe a transparent account of a training curriculum’s failure modes is as scientifically valuable as its successes.

2. Related Work

Classical GRN inference. GENIE3 (Huynh-Thu et al., 2010) and its faster gradient-boosted reimplementation GRNBoost2 (Moerman et al., 2019) frame GRN inference as, for each target gene, a regression of that gene’s expression on all candidate TFs’ expression, with the resulting feature importances read off as edge scores. These methods use only RNA, treat every reference edge as equally reliable, and do not model any graph structure among genes.

Deep generative and graph-based methods. RegDiffusion (Zhu & Slonim, 2024) reframes GRN inference as learning a parameterized adjacency matrix inside a denoising diffusion model over gene expression. GMFGRN (Li et al., 2024) builds a cell-gene bipartite graph and uses graph neural network-based matrix factorization to obtain gene embeddings, from which TF-target compatibility is scored. Both remain RNA-only. scMultiomeGRN (Xu et al., 2025) is, to our knowledge, the closest architecture to ours: it is a supervised graph neural network that combines scRNA-seq and scATAC-seq through modality-specific neighbor aggregators and a cross-modal attention module, and it is the strongest baseline we compare against (Section 9). None of these methods, including scMultiomeGRN, exploit an evidence-confidence hierarchy among training labels; each treats its reference network as a single, flat positive set.

Curriculum learning and catastrophic forgetting. Ordering training examples from easy to hard was proposed by Bengio et al. (2009) and shown to improve convergence and generalization in several settings. A well-documented failure mode of sequential training on ordered data is *catastrophic forgetting* (McCloskey & Cohen, 1989): gradient updates for a new stage overwrite representations useful for earlier stages. Mitigations include regularizing weight changes toward earlier-important parameters, e.g. elastic weight consolidation (Kirkpatrick et al., 2017), and experience replay, which periodically re-introduces earlier-stage examples during later training (Rolnick et al., 2019). We adopt a replay-based mitigation (Section 6) and, notably, find that in our setting replay alone is insufficient without an additional consolidation stage – consistent with reports that replay strength and schedule matter more than its mere presence (Rolnick et al., 2019).

Positioning. MEvD-GRN differs from all of the above along two axes simultaneously: (i) it is trained on a curriculum defined by *experimental evidence confidence* rather than an arbitrary difficulty heuristic, and (ii) its decoder is explicitly role-aware, treating ATAC as a gating precondition on RNA rather than a signal to be concatenated or averaged with it.

3. Problem Formulation

Let $\mathcal{G} = \mathcal{V}_{\text{TF}} \cup \mathcal{V}_{\text{gene}}$ be the gene universe for a given cell type, with $\mathcal{V}_{\text{TF}} \subset \mathcal{V}_{\text{gene}}$ the subset of genes that are annotated as transcription factors. For a directed candidate pair $(i, j) \in \mathcal{V}_{\text{TF}} \times \mathcal{V}_{\text{gene}}$, GRN inference seeks a scoring function $f_{\theta}(i, j) \rightarrow \mathbb{R}$ such that higher scores indicate higher confidence that TF i regulates gene j . The candidate space has size $|\mathcal{V}_{\text{TF}}| \times (|\mathcal{V}_{\text{gene}}| - 1)$, and true regulatory edges are a minuscule fraction of it.

Definition 3.1 (Evidence tiers). Let $E_{\text{loc}}, E_{\text{pert}}, E_{\text{dual}} \subseteq \mathcal{V}_{\text{TF}} \times \mathcal{V}_{\text{gene}}$ denote the sets of positive edges supported by localization, perturbation, and dual evidence, respectively, as curated by SC-MO-GRN-DB for a given cell type. We refer to $(E_{\text{loc}}, E_{\text{pert}}, E_{\text{dual}})$, in that order, as the **evidence hierarchy**: easiest-to-obtain / least specific first, hardest-to-obtain / most specific last.

Crucially, these sets are *not* independently sampled subsets of some universal true edge set – they are nested to varying degrees, because dual-evidence edges are by definition edges that also satisfy the perturbation (and typically the localization) criterion. We quantify this nesting for K562 in Section 4 and use it directly to motivate our curriculum design in Section 6.

4. Data

4.1. SC-MO-GRN-DB and cell-line selection

SC-MO-GRN-DB (Valensi et al., 2026) is, to our knowledge, the only public resource that pairs experimentally-derived reference GRNs with tissue-matched single-cell multiomic data and explicitly labels each reference network’s evidence type. In total it catalogs 30 reference networks (1,616 TFs, 63,512 target genes, 22.4M edges) and 31 single-cell datasets (2.3M cells across 16 tissue types, human and mouse). Only two of these ~ 16 cell types – **K562** (human chronic myelogenous leukemia) and **ESC** (mouse embryonic stem cell) – have reference networks for *all three* evidence tiers; every other cell type in the database has localization evidence only. This paper reports results for **K562**; ESC and the localization-only cell lines are reserved for the zero-shot cross-cell-type transfer study outlined as future work in Section 11.

Table 1. K562 data footprint.

Quantity	Value
scRNA-seq cells	953
scATAC-seq cells (post-QC)	1,126
Final gene universe	22,943
of which TFs	225
Candidate TF $\times$ gene pairs	5,161,950
Co-expression kNN graph ($k=20$)	871,450
TF-candidate graph (top-500/TF)	112,500
Random negative pool	1,000,000

Table 2. K562 evidence-tier edge counts, before and after gene-universe alignment.

Tier	Raw	Aligned
Localization	1,435,721	1,168,069
Perturbation	236,426	230,211
Dual-evidence	29,299	26,618

4.2. K562 data footprint

Tables 1–2 summarize the raw and processed data. The scRNA-seq profile (953 cells) is normalized with counts-per-10k (CP10K) followed by $\log(1 + x)$; the scATAC-seq profile (1,126 cells post-QC, 10x multiome) is converted to a per-gene activity score by multiplying the cell $\times$ peak matrix by a peak $\rightarrow$ gene TSS-proximity incidence matrix (window $\pm 100\text{kb}$). After intersecting RNA genes, ATAC-mapped genes, and every network’s TF/target gene symbols, the final gene universe contains **22,943 genes**, of which **225 are TFs**, giving a candidate space of $225 \times 22,942 = 5,161,950$ directed TF $\rightarrow$ gene pairs.

4.3. The evidence tiers are strongly nested

Figure 1 (left panel) shows the raw sizes of the three tiers; the right panel shows their pairwise overlap. Perturbation overlaps localization by only 9.0% (consistent with the two assays capturing largely different phenomena, as expected from their differing precision/recall profile). Dual-evidence, however, is 100% contained within perturbation and 77.5% contained within localization at K562 – which is expected, since dual-evidence is defined as their intersection, but the *degree* of nesting (essentially total, in the case of perturbation) is large enough that it must be accounted for explicitly in both the splitting protocol (Section 4.4) and the curriculum design (Section 6).

4.4. A global, hierarchy-consistent splitting protocol

Because the tiers are nested rather than disjoint, splitting each tier *independently* into train/validation/test (as is standard practice when tiers are assumed unrelated) does not, by

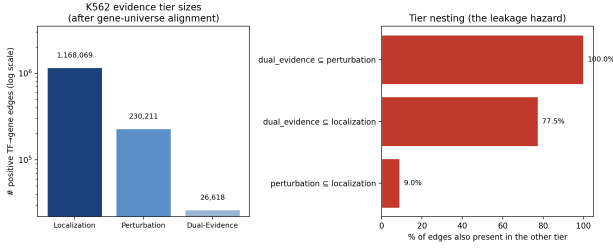


Figure 1. K562 evidence-tier sizes (left) and pairwise nesting fractions (right). Perturbation is a largely disjoint, high-precision complement to localization (9.0% overlap), while dual-evidence is almost entirely a subset of both (100% of dual-evidence $\subseteq$ perturbation, 77.5% $\subseteq$ localization).

Algorithm 1 Global hierarchy-consistent edge splitting

Input: tier edge sets $E_{\text{loc}}, E_{\text{pert}}, E_{\text{dual}}$; ratios $(r_{\text{tr}}, r_{\text{val}}, r_{\text{te}}) = (0.70, 0.15, 0.15)$; seed $U \leftarrow E_{\text{loc}} \cup E_{\text{pert}} \cup E_{\text{dual}}$ pool unique pairs across all tiers
 Shuffle U deterministically with *seed*
 Partition the shuffled U into $(U_{\text{tr}}, U_{\text{val}}, U_{\text{te}})$ by $(r_{\text{tr}}, r_{\text{val}}, r_{\text{te}})$
 $\pi(e) \leftarrow s$ for every $e \in U_s, s \in \{\text{tr}, \text{val}, \text{te}\}$
for each tier $t \in \{\text{loc}, \text{pert}, \text{dual}\}$ **do**
 for each split $s \in \{\text{tr}, \text{val}, \text{te}\}$ **do**
 $\text{split}_t(s) \leftarrow E_t \cap \pi^{-1}(s)$
 end for
end for
Output: per-tier splits $\{\text{split}_t(s)\}$

itself, guarantee that a (TF, gene) pair assigned to one tier’s evaluation split is absent from another tier’s training split – an independent random draw per tier can place the same underlying edge into different splits for different tiers. Since this is exactly the situation created by the nesting quantified in Section 4.3, we instead adopt a splitting protocol that assigns a single, global split label to every unique (TF, gene) pair, once, across the *union* of all evidence tiers, and then defines each tier’s own train/validation/test partition as the intersection of that tier’s edge set with the global assignment.

Because π assigns exactly one split label per unique pair, independent of which tier(s) that pair happens to belong to, Algorithm 1 satisfies a structural non-leakage property by construction:

Proposition 4.1 (Cross-tier split consistency). *Let π be the global assignment produced by Algorithm 1. For any two tiers t, t' and any edge $e \in E_t \cap E_{t'}$, e is assigned to the same split under both tiers: $e \in \text{split}_t(s) \Leftrightarrow e \in \text{split}_{t'}(s)$. Consequently, no edge can be a training positive under one tier while simultaneously being a validation or test positive*

under another tier.

Proposition 4.1 is what licenses the zero-shot evaluation methodology in Section 6: any dual-evidence edge that falls in $\pi^{-1}(\text{val}) \cup \pi^{-1}(\text{test})$ is guaranteed to not have been a training positive under localization or perturbation either, however deeply dual-evidence is nested inside them. We use 5 sampled negatives per positive for training and 5 for evaluation, drawn from a pre-sampled pool of 1,000,000 random non-positive TF $\times$ gene pairs; we verified in early experiments that ranking negatives by chromatin accessibility (rather than sampling them uniformly at random) inflates AUPR through an expression-level artifact unrelated to genuine regulatory signal, and therefore use uniform random negative sampling throughout.

5. Model Architecture

5.1. Design philosophy

RNA and ATAC are not interchangeable evidence for the same underlying quantity; they answer two different biological questions. Whether TF i can regulate gene j depends on (a) whether j ’s locus is *accessible* – an ATAC property, and a *precondition* rather than a graded signal – and (b) whether i and j are transcriptionally *compatible*, e.g. co-expressed across the same cell states – an RNA property. MEvD-GRN reflects this asymmetry structurally: RNA and ATAC are encoded and message-passed in two separate towers that never mix until the very last step, the decoder, where the ATAC representation acts as a multiplicative *gate* on the RNA representation rather than being concatenated or averaged with it.

5.2. Modality encoders

Each gene g is represented by two small feature vectors, built once per cell type from the single-cell data (Section 4): an RNA feature $x_g^{\text{rna}} \in \mathbb{R}^3$ (mean expression, variance, detection rate) concatenated with a 50-dimensional co-expression signature s_g obtained from a truncated SVD of the standardized expression matrix such that $s_i \cdot s_j \approx \text{corr}(g_i, g_j)$ without ever materializing the dense gene $\times$ gene correlation matrix; and an ATAC feature $x_g^{\text{atac}} \in \mathbb{R}^2$ (mean and variance of gene-activity score), with a scalar locus openness feature used later, only at the decoder.

$$h_g^{\text{rna}} = \text{LN}(W_2 \text{Dropout}(\text{GELU}(\text{LN}(W_1 x_g^{\text{rna}})))) \quad (1)$$

$$h_g^{\text{atac}} = \text{LN}(\text{GELU}(W_a x_g^{\text{atac}})) \quad (2)$$

with $W_1 \in \mathbb{R}^{64 \times 3}$, $W_2 \in \mathbb{R}^{128 \times 64}$, $W_a \in \mathbb{R}^{128 \times 2}$, mapping both modalities into a shared 128-d space that is nonetheless kept in *separate tensors* per modality.

5.3. Dual relational GNN towers

Two message-passing graphs, built once from structure only (never from labeled positive edges, to avoid leaking label information into the node embeddings themselves), are shared by both modality towers: a gene–gene co-expression k NN graph ($k = 20$, undirected) capturing co-regulation modules, and a TF→candidate-target graph connecting each TF to its top-500 accessible and co-expressed targets (directionality is used only to construct the graph; the graph is treated as bidirectional for message passing, and directionality is reintroduced only at the decoder). Each tower is a 2-layer relational GraphSAGE (Hamilton et al., 2017): every layer runs one `SAGEConv` per relation and sums the results before a shared LayerNorm + GELU + dropout, so a node’s updated representation mixes both co-expression neighbors and regulatory-candidate neighbors at every layer:

$$h^{(\ell+1)} = \phi\left(\sum_{r \in \mathcal{R}} \text{SAGE}_r^{(\ell)}(h^{(\ell)})\right) \quad (3)$$

where $\mathcal{R} = \{\text{coexpr}, \text{TF-cand}\}$ and $\phi = \text{Dropout} \circ \text{GELU} \circ \text{LN}$, applied identically, but with independent weights, to the RNA tower (h^{rna}) and the ATAC tower (h^{atac}).

5.4. Role-aware decoder

The decoder is where the two modalities finally interact, and it does so asymmetrically by design: ATAC gates RNA, not the reverse.

$$a_j = \sigma(W_{\text{gate}} h_j^{\text{atac}}) \quad (4)$$

$$z_j = h_j^{\text{rna}} \odot a_j \quad (5)$$

$$\text{raw} = (h_i^{\text{rna}})^\top W z_j \quad (6)$$

$$\text{logit} = \text{raw} + w_c \cdot \text{coexpr}(i, j) + w_o \cdot \text{open}_j + b \quad (7)$$

Here $a_j \in [0, 1]^{128}$ is a per-dimension *accessibility gate* derived from the target’s ATAC embedding; z_j is the target’s resulting “regulatable” RNA state; raw is an asymmetric bilinear TF→target compatibility score; and $\text{coexpr}(i, j) = s_i \cdot s_j$ is the co-expression-signature dot product, while open_j is the locus-openness scalar from Section 4. Both additive terms let the decoder fall back on direct co-expression and raw accessibility evidence even when the learned bilinear term is uncertain. The model outputs logits; sigmoid is applied only at the loss/eval boundary.

5.5. Parameter budget

Table 3 breaks down the 306,691 trainable parameters by module. For reference, this is roughly **13× smaller** than a single `scMultiomeGRN` model (4,077,914 parameters; Section 8). Figure 2 shows the full architecture.

Table 3. MEvD-GRN parameter count by module.

Module	Parameters
RNA encoder (MLP, $3 \rightarrow 64 \rightarrow 128$)	8,960
ATAC encoder (linear, $2 \rightarrow 128$)	640
GNN _{RNA} tower (2-layer rel. SAGE)	132,096
GNN _{ATAC} tower (2-layer rel. SAGE)	132,096
Role-aware decoder	32,899
Total	306,691

5.6. Architectural ablation hooks

Three flags let us isolate the contribution of each design choice experimentally (Section 10): `integration=gated/concat` replaces the role-aware decoder with a single fused embedding and a classical asymmetric bilinear decoder; `use_atac=False` zeroes the ATAC branch entirely (RNA-only); and `use_gnn=False` feeds encoder outputs directly to the decoder, removing message passing altogether.

6. Training Strategy

6.1. A two-stage evidence-tier curriculum

MEvD-GRN is trained sequentially: Stage 1 on the localization tier (30 epochs, learning rate 10^{-3}), then Stage 2 on the perturbation tier (15 epochs, learning rate 3×10^{-4}), both with the full network unfrozen. We deliberately *exclude* the dual-evidence tier from training altogether and reserve it as a zero-shot generalization benchmark, evaluated only at inference time after Stage 2. This is not merely a convenient simplification; it is motivated directly by the nesting structure quantified in Section 4.3. Because dual-evidence is 100% contained in perturbation and 77.5% contained in localization at K562, a model that is fine-tuned directly on dual-evidence edges would, for the large majority of those edges, be fitting labels for (TF, gene) pairs whose regulatory relationship it has already substantially been exposed to in an upstream tier – so a good score on a held-out *split* of dual-evidence edges would conflate genuine generalization to the highest-confidence tier with memorization of a near-duplicate of what was already trained on. Evaluating a model that has *never seen a single dual-evidence gradient update* sidesteps this ambiguity entirely: by Proposition 4.1, every dual-evidence edge used at evaluation time is guaranteed to not have been a positive example under *any* tier’s training split, making the resulting score an unambiguous measurement of transfer to the highest-confidence evidence tier rather than a measurement of fine-tuning capacity. We treat this zero-shot dual-evidence score as the paper’s headline generalization metric (Section 9).

Evaluation follows a matching convention

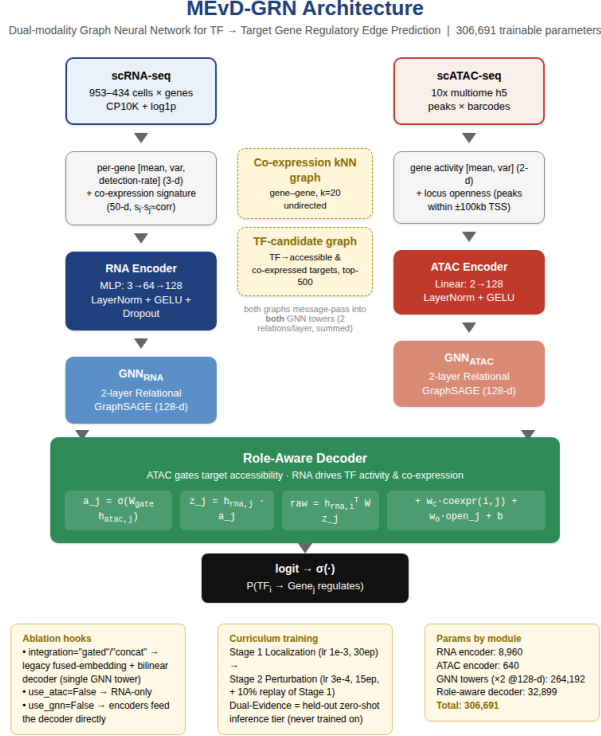


Figure 2. MEvD-GRN architecture. RNA and ATAC signals are encoded and message-passed in separate towers over two shared structural graphs (co-expression kNN and TF-candidate), and combine only at the role-aware decoder, where the ATAC embedding gates the RNA embedding before an asymmetric bilinear compatibility score is computed.

(EVALEDGESFORTIER): tiers that were trained on (localization, perturbation) are scored on their *test* split only, exactly as in standard practice; the held-out dual-evidence tier, having no training split to speak of, is scored on its *validation* + *test* splits combined, since both are equally unseen and combining them yields a larger, less noisy estimate.

6.2. Hard-negative mining

For each training stage, half of the sampled negative pairs are drawn uniformly at random from the global negative pool (Section 4); the other half are *hard negatives* – pairs that are positive in an easier tier of the hierarchy but not in the current training tier. This forces the decoder to distinguish “binds but does not (yet, functionally) regulate” from “binds and regulates,” rather than learning a shortcut that conflates the two notions.

Table 4. Catastrophic forgetting of the localization tier after Stage 2, and how much of it different mitigations recover. AUROC < 0.5 indicates the model’s ranking is worse than random on the forgotten tier.

Configuration	Localization AUROC
No replay (2-stage)	0.290
Replay, 2-stage (shipped default)	0.326
Replay, full 3-stage + consolidation	0.736

6.3. Memory replay: mechanism and an honest quantitative result

During Stage 2, a small fraction (`replay_weight=0.1`, i.e. 10% of the current batch size) of Stage 1’s localization positives are re-injected into every batch at reduced loss weight (0.1), in the spirit of experience replay (Rolnick et al., 2019). We report the effect of this mechanism honestly rather than presenting it as a solved problem. Table 4 shows localization AUROC measured *after* Stage 2 completes, under three conditions:

Replay alone, within the two-stage curriculum actually used by the shipped model, provides only a modest recovery (0.290 → 0.326) and does not resolve the underlying problem: post-Stage-2 localization AUROC remains *below random* (< 0.5). The much larger recovery (0.736) is achieved only by a qualitatively different recipe – a full three-stage curriculum that adds a frozen-encoder consolidation pass with replay over all tiers – which we study as an ablation (`with_replay`, Section 10) rather than adopt as the shipped default, precisely because isolating *why* replay alone under-performs is itself an informative result (Section 10.1).

6.4. Optimization details

We use AdamW (weight decay 10^{-4}), gradient-norm clipping at 1.0, batch size 8,192, and cosine learning-rate decay within each stage. The loss is a class-balanced weighted binary cross-entropy with $\text{pos_weight} = n_{\text{neg}}/n_{\text{pos}}$ recomputed per stage. We use early stopping on validation AUPR with a patience of 10 checks, evaluating every 5 epochs. End-to-end training of the shipped 2-stage model takes **16.6 minutes** on a single consumer GPU (RTX 4060).

7. Evaluation Protocol

We report four metrics, computed per evidence tier: **AUPR**, the area under the precision-recall curve, our primary metric because it is far more sensitive than AUROC to the severe class imbalance inherent to link prediction over a candidate space where true edges are $\ll 1\%$ of all pairs; **AUROC**, the area under the ROC curve, retained as a secondary metric

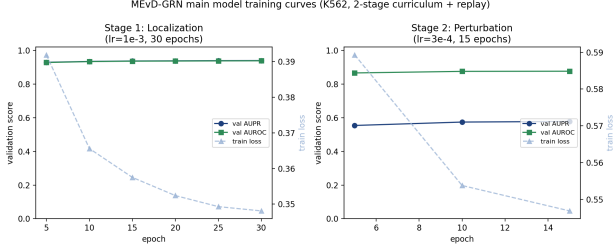


Figure 3. K562 training curves for the shipped 2-stage model. Stage 1 converges smoothly to validation AUPR/AUROC ≈ 0.94 on localization. Stage 2’s own validation curve (perturbation, AUROC 0.876) looks healthy in isolation – the forgetting reported in Table 4 is invisible from Stage 2’s own curve and appears only when Stage 1’s tier is re-evaluated afterward.

because it exposes below-random ranking that AUPR alone can mask (as in Table 4); **Early Precision (EP)**, the precision restricted to the top- k scored candidate pairs with k set to the number of true positives, which measures how useful the very top of a ranked candidate list is in practice; and **EPR**, EP normalized by the random-guessing base rate over the *full* bipartite candidate space ($225 \times 22,942 = 5,161,950$ pairs, following the BEELINE enrichment convention). Because the candidate space is enormous relative to the positive set, EPR values of 10^1 – 10^3 are expected even when the corresponding AUPR looks numerically modest.

8. Baselines

We compare against four baselines, all evaluated on the *identical* splits and evaluation convention as MEvD-GRN (Algorithm 1, Section 6.1), for a fully apples-to-apples comparison. **GRNBoost2** (Moerman et al., 2019) is the field-standard tree-ensemble co-expression method (RNA-only, non-parametric). **RegDiffusion** (Zhu & Slonim, 2024) is a diffusion-based GRN inference model (RNA-only); we restrict its input gene set to highly-variable genes and TFs to avoid zero-variance instabilities introduced by its internal normalization. **GMF-GAE** is a graph-autoencoder baseline over the co-expression k NN graph, standing in for the GMFGRN family (Li et al., 2024) (RNA-only). **scMultiomeGRN** (Xu et al., 2025) is the strongest baseline: a supervised multi-omic GNN with cross-modal attention, at 4,077,914 parameters *per trained tier* (it fits a dedicated model per tier rather than sharing weights across a curriculum, so it never experiences catastrophic forgetting by construction). Its original formulation targets TF $\leftrightarrow$ TF interactions; we adapt it to our TF $\rightarrow$ all-genes benchmark, decoding edges sparsely so the full $|\mathcal{V}_{\text{TF}}| \times |\mathcal{V}_{\text{gene}}|$ score matrix is never materialized. We train it for 100 epochs per tier (an exploratory budget; its published default is 2,000 epochs – see Section 11).

Table 5. MEvD-GRN main results on K562. Localization/perturbation are scored on their test split (trained tiers); dual-evidence is scored on validation+test, zero-shot (never trained on).

Tier	AUPR	AUROC	EP	EPR
Localization [†]	0.420	0.326	0.425	12.5
Perturbation	0.575	0.876	0.536	80.2
Dual-evidence [‡]	0.558	0.838	0.419	272.5

[†]affected by catastrophic forgetting, Section 6.3. [‡]zero-shot, never trained on.

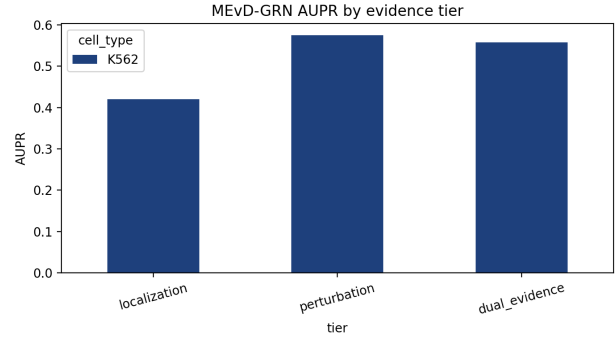


Figure 4. MEvD-GRN AUPR by evidence tier on K562.

9. Results

9.1. Main results

Table 5 reports MEvD-GRN’s performance on all three evidence tiers. The headline result is the dual-evidence row: evaluated in a strict zero-shot setting (Section 6.1), the model reaches AUPR 0.558 / AUROC 0.838 on a tier it never received a single gradient update from – evidence of genuine transfer rather than memorization, a claim made rigorous by Proposition 4.1. The localization row, however, carries the caveat quantified in Section 6.3: AUROC 0.326 is below random, the direct, visible cost of catastrophic forgetting.

9.2. Comparison against baselines

Table 6 compares AUPR across all five methods on all three tiers. scMultiomeGRN dominates localization (0.831 AUPR), which is expected: it fits a dedicated model per tier with no sequential fine-tuning and hence no forgetting, at roughly $13\times$ the parameter count of a single MEvD-GRN model. MEvD-GRN wins clearly on perturbation and, more importantly, on dual-evidence *under true zero-shot transfer* – despite never training on that tier and using a fraction of the parameters. The RNA-only classical and diffusion baselines (GRNBoost2, RegDiffusion, GMF-GAE) trail on every tier,

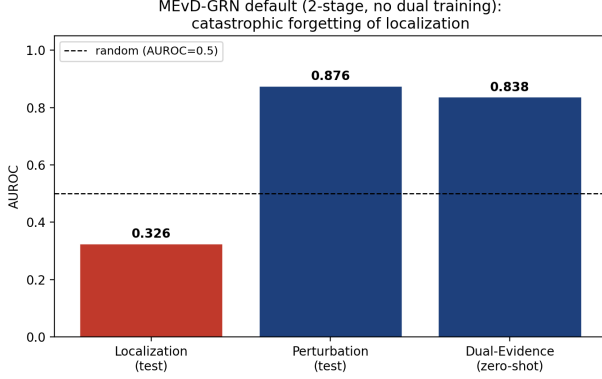


Figure 5. Localization AUROC over the curriculum: Stage 1 reaches ≈ 0.94 , then collapses to 0.326 after Stage 2 – below the 0.5 random-ranking line (dashed).

Table 6. AUPR comparison across methods and tiers, K562. Dual-evidence AUPR for MEvD-GRN is a zero-shot score; all other entries are in-distribution (trained/fit on that tier).

Method	Loc.	Pert.	Dual
MEvD-GRN (ours)	0.420	0.575	0.558
scMultiomeGRN	0.831	0.423	0.384
GRNBoost2	0.535	0.206	0.173
RegDiffusion	0.546	0.185	0.165
GMF-GAE	0.526	0.218	0.160

consistent with none of them exploiting chromatin accessibility or the evidence-tier structure.

9.3. Head-to-head against scMultiomeGRN under alternative curricula

Because scMultiomeGRN is the only baseline that is itself a supervised multi-omic GNN, we examine it more closely, alongside two *alternative* MEvD-GRN training schedules already characterized by the ablation study (Section 10): *all_at_once* (localization and perturbation trained jointly, no sequencing at all) and *with_replay* (the full three-stage curriculum with consolidation from Table 4). Table 7 shows that no single *already-tested* MEvD-GRN configuration wins all three tiers outright, but the choice of curriculum entirely determines which tiers are won: the shipped default trades localization for a dual-evidence win, while *with_replay* very nearly ties scMultiomeGRN on both localization and perturbation while beating it by 2.4 $\times$ on dual-evidence – using roughly 27 $\times$ fewer parameters than the three dedicated per-tier scMultiomeGRN models such a sweep would otherwise require.

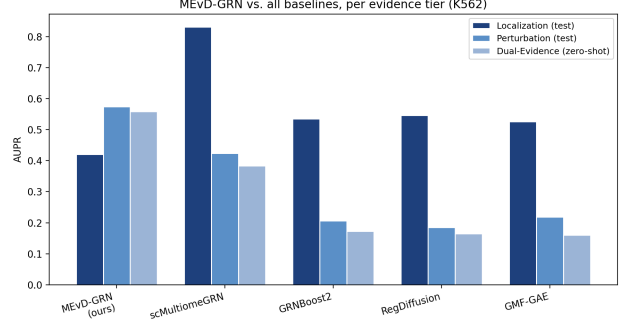


Figure 6. Full per-tier AUPR comparison across all five methods.

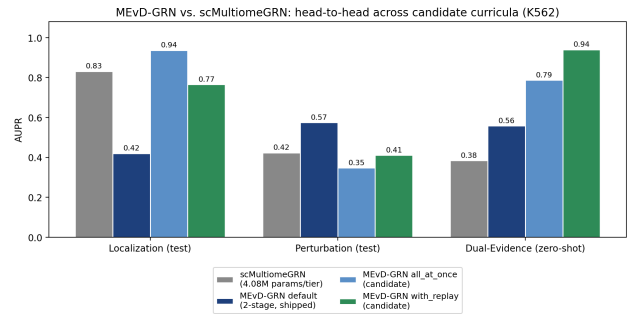


Figure 7. MEvD-GRN (three curricula) vs. scMultiomeGRN, AUPR by tier.

10. Ablation Study

We run 10 configurations under the identical splitting and evaluation protocol (Section 7) to isolate the contribution of the curriculum, the GNN backbone, the fusion strategy, and each modality. Table 8 reports AUPR by tier for every configuration; AUROC/EP/EPR for all configurations are given in Appendix A.

10.1. Discussion of ablation findings

Sequential ordering is not, by itself, beneficial once forgetting is accounted for. Comparing *full_curriculum* against *all_at_once* isolates the effect of *sequencing* alone (both train on the same two tiers, localization and perturbation; the only difference is joint vs. sequential optimization): *all_at_once* dominates on dual-evidence zero-shot generalization (0.788 vs. 0.546 AUPR) and on localization (0.937 vs. 0.410), while trading off a smaller amount of perturbation performance (0.348 vs. 0.569). This shows that the sequential curriculum’s advantage, if any, is more than offset by the forgetting it introduces (Section 6.3) in the present configuration – a genuinely negative result for naive sequential ordering that we report because it directly

Table 7. Head-to-head vs. scMultiomeGRN across curriculum choices (K562, AUPR / AUROC). scMultiomeGRN uses a dedicated 4.08M-parameter model per tier; all three MEvD-GRN rows use the *same* single 306,691-parameter model, only the training schedule differs.

Method (parameters)	Localization	Perturbation	Dual-evidence
scMultiomeGRN (4.08M $\times$ up to 3 dedicated models)	0.831 / 0.865	0.423 / 0.828	0.384 / 0.825
MEvD-GRN — shipped default, 2-stage (306,691 params)	0.420 / 0.326	0.575 / 0.876	0.558 / 0.838
MEvD-GRN — <code>all_at_once</code> (same params)	0.937 / 0.928	0.348 / 0.739	0.788 / 0.943
MEvD-GRN — <code>with_replay</code> (same params)	0.765 / 0.736	0.411 / 0.702	0.940 / 0.990

Table 8. Ablation study, K562 (AUPR by tier). `full_curriculum` is the shipped 2-stage default, reproduced here as the reference row. `dual_only` and `with_replay` are explicit exceptions that do train on dual-evidence, included to bound the comparison.

Configuration	Trained on	Localization	Perturbation	Dual-evidence
<code>full_curriculum</code> (reference)	loc $\rightarrow$ pert, sequential	0.410	0.569	0.546
<code>loc_only</code>	localization only	0.940	0.187	0.663
<code>pert_only</code>	perturbation only	0.407	0.496	0.485
<code>dual_only</code> *	dual-evidence only	0.779	0.197	0.479
<code>all_at_once</code>	loc + pert, simultaneous	0.937	0.348	0.788
<code>rna_only</code>	loc $\rightarrow$ pert, ATAC zeroed	0.414	0.540	0.543
<code>gated_fusion</code>	loc $\rightarrow$ pert, legacy fusion	0.419	0.588	0.546
<code>concat_fusion</code>	loc $\rightarrow$ pert, plain concat	0.421	0.595	0.555
<code>no_gnn</code>	loc $\rightarrow$ pert, no message passing	0.498	0.341	0.241
<code>with_replay</code> *	full 3-stage + replay	0.765	0.411	0.940

*deliberately trains on dual-evidence and is excluded from the zero-shot claim of Section 9.3; included here for completeness.

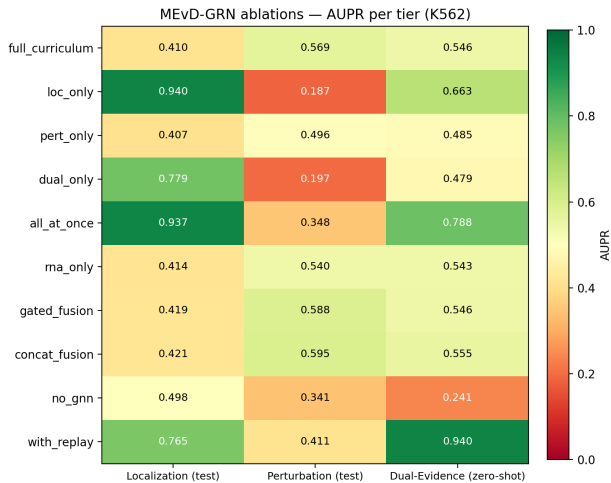


Figure 8. Ablation heatmap: AUPR by configuration $\times$ tier.

qualifies the curriculum-learning hypothesis motivating this work.

Replay with a consolidation pass recovers most of the deficit. `with_replay`, which adds a third, frozen-encoder consolidation stage with replay over all tiers on top of the sequential curriculum, achieves the best dual-evidence score of any configuration (0.940 AUPR) and substantially recov-

ers localization (0.765, vs. 0.410 for the plain sequential curriculum), at some cost to perturbation (0.411 vs. 0.569). Combined with the `all_at_once` result, this suggests two independently viable routes to a stronger default recipe than the currently shipped one (Section 11).

The GNN backbone is the single highest-value architectural component. `no_gnn` is the worst configuration on every tier (localization 0.498, perturbation 0.341, dual 0.241), confirming that relational message passing over the co-expression and TF-candidate graphs contributes substantially more than any other architectural choice we ablate.

Fusion style is a second-order effect at the point-metric level. Role-aware, gated, and concatenation fusion land within ≈ 0.02 – 0.03 AUPR of one another under the shipped curriculum (Table 8, rows `full_curriculum`/`gated_fusion`/`concat_fusion`); the role-aware decoder’s advantage is currently more interpretive/biologically-motivated (Section 5) than a large raw-metric win, and we flag this honestly as an open question rather than overstate it.

ATAC’s marginal contribution is currently small. `rna_only` (ATAC branch zeroed) performs almost identically to the full model (0.414/0.540/0.543 vs. 0.410/0.569/0.546), suggesting that under the current coarse, gene-level (mean/variance) ATAC featurization, chromatin accessibility is not yet contributing information

that the RNA co-expression signature does not already capture. We view this as an artifact of the featurization rather than a fundamental property of the data, and flag it as a priority for follow-up work (Section 11).

11. Limitations and Future Work

We highlight four limitations, in order of priority. **(1) The shipped default curriculum is not the strongest available configuration.** Section 10.1 shows that both `all_at_once` and `with_replay` outperform the shipped 2-stage-with-light-replay default on the majority of tiers; promoting one of these to the new default, and tuning the replay schedule (weight, source stages, consolidation length) that was not swept in this study, is the most direct next step. **(2) Only K562 has been trained and evaluated end-to-end.** ESC is the only other cell type with all three evidence tiers, and localization-only cell types (used for a strictly harder zero-shot cross-cell-type transfer test) are still being processed; no cross-cell-type number yet exists, and every result in this paper is K562-specific. **(3) ATAC’s measured contribution is currently small** (Section 10.1); whether this reflects a genuine property of K562 regulation or the coarseness of a gene-level, mean/variance ATAC featurization (as opposed to, e.g., peak-level or motif-aware features) is an open question. **(4) scMultiomeGRN was trained at a reduced, 100-epoch exploratory budget** rather than its published default of 2,000 epochs; the comparison in Section 9 is protocol-fair (identical splits and evaluation convention) but not yet at the baseline’s full training budget, and we intend to re-run it at full budget before treating Table 6 as final.

12. Conclusion

We introduced MEvD-GRN, a compact, role-aware dual-modality graph neural network trained on an evidence-tier curriculum derived directly from how regulatory edges were experimentally discovered rather than from an arbitrary difficulty heuristic. Held out entirely from training and evaluated zero-shot, the highest-confidence dual-evidence tier is where MEvD-GRN is most convincing, exceeding a baseline with more than $13\times$ its parameter count. At the same time, our ablation study surfaces a genuine and quantified weakness – catastrophic forgetting under the shipped sequential curriculum – and identifies two concrete, already-tested alternative training recipes that substantially close the resulting gap. We view the combination of a positive headline result and an honestly quantified failure mode, together with the formal non-leakage guarantee underlying our zero-shot evaluation methodology (Proposition 4.1), as the paper’s main contribution to how confidence-tiered biological reference data should be used for GRN inference going forward.

Impact Statement

This paper presents work on computational inference of gene regulatory networks from single-cell multi-omic data, intended to advance basic biological understanding and, downstream, target identification for therapeutic research. All data used are from a public, de-identified reference database of established human and mouse cell lines (Valensi et al., 2026); no new experimental or clinical data were collected. We are not aware of any specific societal risk beyond those generally associated with advancing computational biology and machine learning methodology.

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A. Full Ablation Metrics

Table 9 reports all four metrics (AUPR, AUROC, EP, EPR) for every ablation configuration and every tier, complementing the AUPR-only view in Table 8.

Table 9. Full ablation results, K562, all metrics.

Configuration	Tier	AUPR	AUROC	EP	EPR
full_curriculum	localization	0.410	0.298	0.415	12.2
full_curriculum	perturbation	0.569	0.875	0.531	79.6
full_curriculum	dual_evidence	0.546	0.837	0.421	273.4
loc_only	localization	0.940	0.939	0.870	25.6
loc_only	perturbation	0.187	0.365	0.150	11.2
loc_only	dual_evidence	0.663	0.799	0.650	422.7
pert_only	localization	0.407	0.282	0.384	5.7
pert_only	perturbation	0.496	0.825	0.465	69.7
pert_only	dual_evidence	0.485	0.797	0.387	251.8
dual_only	localization	0.779	0.761	0.718	10.6
dual_only	perturbation	0.197	0.501	0.195	14.6
dual_only	dual_evidence	0.479	0.778	0.514	666.4
all_at_once	localization	0.937	0.928	0.865	25.5
all_at_once	perturbation	0.348	0.739	0.328	49.1
all_at_once	dual_evidence	0.788	0.943	0.701	455.4
rna_only	localization	0.414	0.307	0.415	12.2
rna_only	perturbation	0.540	0.860	0.499	74.7
rna_only	dual_evidence	0.543	0.825	0.415	269.9
gated_fusion	localization	0.419	0.323	0.426	12.6
gated_fusion	perturbation	0.588	0.882	0.549	82.2
gated_fusion	dual_evidence	0.546	0.838	0.424	275.6
concat_fusion	localization	0.421	0.330	0.433	12.8
concat_fusion	perturbation	0.595	0.886	0.556	83.2
concat_fusion	dual_evidence	0.555	0.848	0.444	288.2
no_gnn	localization	0.498	0.473	0.524	15.4
no_gnn	perturbation	0.341	0.710	0.369	55.3
no_gnn	dual_evidence	0.241	0.666	0.261	169.5
with_replay	localization	0.765	0.736	0.697	20.5
with_replay	perturbation	0.411	0.702	0.372	55.7
with_replay	dual_evidence	0.940	0.990	0.874	1134.3

B. Environment and Reproducibility

Python 3.10, PyTorch ≥ 2.1 , PyTorch Geometric ≥ 2.4 , scikit-learn, pandas, scipy, matplotlib; optional `arboreto` (GRN-Boost2) and `regdiffusion` packages for baselines. All experiments were run on a single consumer GPU (RTX 4060); the full model has under 400K parameters, so a roughly 1:10 GPU:CPU utilization ratio is sufficient. All random splits use a fixed seed (Algorithm 1) for reproducibility.